

Graph Puzzles IV.1-preview: Factorisations of hypohamiltonian and permutation snarks

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ABSTRACT: Every hypohamiltonian snark and every permutation snark has a unique multiset of cyclically 5-edge-connected factors of the same class under successive decompositions along cycle-separating 4-edge-cuts. We prove that hypohamiltonicity passes from a dot product to both of its snark factors. Together with the factorisation theorem for bicritical snarks, this answers a question of Goedgebeur and Zamfirescu. For permutation snarks, we extend the atom argument of Chladný and Škoviera using colourable cut multipoles and closure under their canonical completions. The two conclusions also hold within the class of hypohamiltonian permutation snarks.

1 Introduction

The dot product combines two snarks across a 4-edge-cut. Reversing this operation repeatedly expresses a snark in terms of factors with higher cyclic connectivity. Chladný and Škoviera [CS10]¹ proved that a bicritical snark has a unique multiset of cyclically 5-edge-connected bicritical factors. Their theorem identifies the isomorphism types of the factors and their multiplicities, independently of the cuts chosen during the decomposition.

A graph is *hypohamiltonian* if it is non-Hamiltonian and every single-vertex deletion is Hamiltonian. Hypohamiltonian snarks are bicritical. Goedgebeur and Zamfirescu [GZ18]² asked whether every hypohamiltonian snark of cyclic connectivity four is a dot product of two hypohamiltonian snarks [GZ18, Problem 4.3]. We prove that both snark factors of any hypohamiltonian dot product are hypohamiltonian. The unique factorisation theorem for bicritical snarks then gives the following result.

Theorem 1.1. *Every hypohamiltonian snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected hypohamiltonian snark factors. Every intermediate factor in a decomposition along cycle-separating 4-edge-cuts is hypohamiltonian.*

A *permutation 2-factor* consists of two induced cycles. A snark admitting such a 2-factor is a *permutation snark*. Its two cycles have

¹ Miroslav Chladný and Martin Škoviera. *Factorisation of snarks*. Electronic Journal of Combinatorics 17(1) (2010), R32. [doi:10.37236/304](https://doi.org/10.37236/304).

² Jan Goedgebeur and Carol T. Zamfirescu. *On hypohamiltonian snarks and a theorem of Fiorini*. Ars Mathematica Contemporanea 14(2) (2018), 227–249. [doi:10.26493/1855-3974.1176.9b7](https://doi.org/10.26493/1855-3974.1176.9b7).

equal odd length, and their complement is a perfect matching between them. Máčajová and Škoviera [MS19]³ proved that every cycle-separating 4-edge-cut of a permutation snark determines two smaller permutation snarks [MS19, Section 2 and Theorem 2.1].

Permutation snarks are conjectured to be bicritical; see Máčajová and Škoviera [MS25]⁴, Section 1. Together with the factorisation theorem for bicritical snarks, this conjecture makes it natural to ask whether permutation snarks also have unique factorisations. We use the closure property above to prove the following result.

Theorem 1.2. *Every permutation snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected permutation snark factors. Every intermediate factor in a decomposition along cycle-separating 4-edge-cuts is a permutation snark.*

Permutation snarks include non-hypohamiltonian graphs. There is a 26-vertex example obtained as a successive dot product of three Petersen graphs, and there is also a 50-vertex example with cyclic connectivity five. Brinkmann, Goedgebeur, Hägglund and Markström [BGHM13]⁵ enumerated 64 permutation snarks on 26 vertices [BGHM13, Table 3]. Thus the hypohamiltonian permutation snarks form a proper subclass of the permutation snarks, also within the cyclically 5-edge-connected class. The two factorisation theorems apply to the same completed factors at each cut, giving the following result for their intersection.

Corollary 1.3. *Every hypohamiltonian permutation snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected hypohamiltonian permutation snark factors. Every intermediate factor has both properties.*

2 Dot products and cut multipoles

All graphs are finite. We use *snark* for a simple cubic graph of girth at least five, cyclically 4-edge-connected, and with chromatic index four. A *cycle* is a connected 2-regular subgraph. For $X \subseteq V(G)$, write $\partial_G X$ for the edges with exactly one end in X . The cut is *cycle-separating* when both $G[X]$ and $G[V(G) \setminus X]$ contain a cycle. We write $\lambda_c(G)$ for cyclic edge-connectivity, $g(G)$ for girth, and $|G|$ for $|V(G)|$.

2.1 The dot product

Let A and B be disjoint snarks. Choose independent edges ab, cd of A and an edge xy of B , with

$$N_B(x) = \{a', b', y\}, \quad N_B(y) = \{c', d', x\}.$$

³ Editá Máčajová and Martin Škoviera. *Permutation snarks of order 2 (mod 8)*. Acta Mathematica Universitatis Comenianae 88(3) (2019), 929–934. [journal version](#).

⁴ Editá Máčajová and Martin Škoviera. *Are there any permutation snarks on 6 (mod 8) vertices?* Procedia Computer Science 273 (2025), 163–168. [doi:10.1016/j.procs.2025.10.294](#).

⁵ Gunnar Brinkmann, Jan Goedgebeur, Jonas Hägglund and Klas Markström. *Generation and properties of snarks*. Journal of Combinatorial Theory, Series B 103(4) (2013), 468–488. [doi:10.1016/j.jctb.2013.05.001](#).

The four vertices a', b', c', d' are distinct. Put

$$L = A - \{ab, cd\}, \quad R = B - \{x, y\},$$

where edges are deleted from A and vertices from B . The dot product with this attachment is

$$G = A \cdot B = L \cup R + \{aa', bb', cc', dd'\}. \quad (1)$$

The four joining edges form the *bond* F . Its two couples are $\{aa', bb'\}$ and $\{cc', dd'\}$. This is the convention of [CS10, Section 2] and [GZ18, Section 2]. The order satisfies

$$|G| - 2 = (|A| - 2) + (|B| - 2). \quad (2)$$

A cycle-separating 4-edge-cut in a snark is a matching. Indeed, if a vertex is incident with at least two cut edges, moving it to the other shore reduces the size of the cut and preserves a cycle in each shore. The resulting cut has size at most three, contrary to cyclic 4-edge-connectivity. Thus each shore has four distinct boundary vertices.

2.2 Canonical completions

Disconnecting the four edges of a cut leaves a 4-pole on each shore: a cubic multipole with four semiedges. A proper 3-edge-colouring of a multipole assigns colours to its ordinary edges and semiedges, with three distinct colours incident at each vertex. We use the multipole terminology of [CS10, Section 3].

Up to a permutation of colours, the Parity Lemma permits four types of boundary colouring:

$$1111, \quad 1122, \quad 1212, \quad 1221.$$

Every colourable 4-pole admits at least two of these types. If two colourable 4-poles join to a snark, their sets of types are disjoint, so each admits exactly two. Such a pole is *colour-open*. Its four semiedges have a unique partition into two couples. In the *isochromatic* case, the semiedges of each couple receive the same colour in every colouring. In the *heterochromatic* case, the semiedges of each couple receive distinct colours in every colouring. The two poles of the cut have opposite types, and their couples agree across the cut [CS10, Propositions 3.1–3.5].

The *canonical snark completion* of a heterochromatic pole joins the semiedges within each couple. For an isochromatic pole, it attaches each couple to a new vertex and joins the two new vertices. These completions are unique up to isomorphism. They recover the two

factors in (1), with the heterochromatic completion playing the role of A and the isochromatic completion the role of B .

A *composition chain* of G is a multiset obtained from $\{G\}$ by successively replacing a member with its two canonical factors at a cycle-separating 4-edge-cut. A chain is *terminal* when every member is cyclically 5-edge-connected. Two chains are *equivalent* if their members can be matched, including multiplicities, by graph isomorphisms. We use the terminology of [CS10, Section 7].

3 Hypohamiltonian factors

We first record the boundary restrictions imposed by the non-Hamiltonicity of the two factors.

Lemma 3.1. *In (1), let A and B be non-Hamiltonian graphs. A spanning path in L with terminal ends joins different pairs among $\{a, b\}$ and $\{c, d\}$. A spanning path in R with terminal ends joins a' to b' or c' to d' . Two disjoint paths spanning L , with ends a, b, c, d , pair a with b and c with d . Two disjoint paths spanning R , with ends a', b', c', d' , pair a' with b' and c' with d' .*

Proof. A spanning a – b path in L closes to a Hamiltonian cycle of A when ab is restored; the same holds for c, d . A spanning path in R whose ends belong to different pairs closes through x, y to a Hamiltonian cycle of B .

For a spanning pair of paths in L , either cross pairing of the four ends becomes one Hamiltonian cycle after restoring ab and cd . For a spanning pair in R , either cross pairing becomes one Hamiltonian cycle after adjoining $a'xb'$ and $c'yd'$. The remaining pairing is the one stated in each case. $\square$

Lemma 3.2. *If a dot product $G = A \cdot B$ of snarks is hypohamiltonian, then both A and B are hypohamiltonian.*

Proof. Use the notation of (1). A Hamiltonian cubic graph is 3-edge-colourable, so both factors are non-Hamiltonian. A Hamiltonian cycle in any vertex-deleted subgraph of G meets F in two or four edges. With two crossings, its restriction to the undeleted shore is a spanning path; with four, that restriction consists of two disjoint spanning paths.

Fix $v \in V(A)$ and take a Hamiltonian cycle C in $G - v$. If C uses two bond edges, lemma 3.1 shows that its spanning path in R has ends a', b' or c', d' . Replace the whole segment through R , including its bond edges, with ab or cd , respectively. If C uses all four bond edges, its two spanning paths in R pair a' with b' and c' with d' . Replace

these two segments with ab and cd . Each replacement preserves the cyclic order of the remaining vertices of C . The resulting cycle spans $A - v$. Thus $A - v$ is Hamiltonian for every $v \in V(A)$.

Now fix $w \in V(B) \setminus \{x, y\}$ and take a Hamiltonian cycle in $G - w$. With two bond edges, the spanning path in L joins different terminal pairs. For example, a path from a to c is replaced, together with its bond edges, by $a'xc'$. The other choices give the corresponding path through x, y . With four bond edges, the two spanning paths in L pair a with b and c with d . Replace them, together with their bond edges, by $a'xb'$ and $c'yd'$. The resulting cycle spans $B - w$ in either case.

It remains to recover Hamiltonian cycles after deleting x or y . Take a Hamiltonian cycle of $G - a$. The missing bond edge aa' forces exactly two crossings. Its spanning path in R has ends c', d' by lemma 3.1. Adjoining $c'yd'$ gives a Hamiltonian cycle of $B - x$. A Hamiltonian cycle of $G - c$ similarly has a spanning $a'-b'$ path in R ; adjoining $a'xb'$ gives a Hamiltonian cycle of $B - y$. Hence both factors are hypohamiltonian. $\square$

A snark is *bicritical* if deleting any two distinct vertices yields a 3-edge-colourable graph. Hypohamiltonian snarks have this property [GZ18, Section 1]. For distinct u, v , deleting v from a Hamiltonian cycle of $G - u$ leaves a spanning path in $G - \{u, v\}$. Colour the path alternately with colours 1, 2. The remaining edges form a matching and receive colour 3.

Theorem 1.1. *Every hypohamiltonian snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected hypohamiltonian snark factors. Every intermediate factor in a decomposition along cycle-separating 4-edge-cuts is hypohamiltonian.*

Proof. Let G be hypohamiltonian and let S be any cycle-separating 4-edge-cut. By bicriticality and [CS10, Theorem 7.1], S determines a dot product $G = A \cdot B$ of smaller bicritical snarks. By lemma 3.2, both factors are hypohamiltonian. Repeating the argument keeps every factor hypohamiltonian.

Each decomposition replaces a graph with two smaller graphs, so every chain has a terminal refinement. Its members are cyclically 5-edge-connected hypohamiltonian snarks. The bicritical unique-factorisation theorem [CS10, Theorem C, or Theorem 10.3] identifies any two such terminal chains up to isomorphism and order, including multiplicities. $\square$

In particular, every hypohamiltonian snark of cyclic connectivity four is a dot product of two hypohamiltonian snarks. This gives an affirmative answer to [GZ18, Problem 4.3].

4 A factorisation criterion

The permutation result uses the atom argument for bicritical snarks [CS10, Sections 8–10]. We formulate the argument with the closure hypotheses that enter its proof.

Let $\mathcal{C}$ be a class of snarks closed under isomorphism. Assume that every cycle-separating 4-edge-cut in a member of $\mathcal{C}$ has two colourable shore poles, and that both canonical completions belong to $\mathcal{C}$. For a cyclic fragment X , write $\tilde{X}$ for the completion of its shore pole and $G/\tilde{X}$ for the completion of the complementary pole.

Suppose $\lambda_c(G) = 4$. A *cyclic fragment* is an induced shore of a cycle-separating 4-edge-cut, and an *atom* is a cyclic fragment minimal under inclusion. Since $4 < g(G)$, the atom theorem of Nedela and Škoviera [NS95]⁶ places every atom wholly on one side of every cycle-separating 4-edge-cut; see also [CS10, Theorem 7.2]. The cut $\partial_G X$ of an atom is *atomic*. A cycle-separating cut with a shore $X \cup \{u, v\}$, where u, v are adjacent outside vertices each joined once to X , is *quasiatomic*. These are the cuts associated with X .

Lemma 4.1. *Let $G \in \mathcal{C}$ have cyclic connectivity four and let X be an atom. A cut associated with X gives factors isomorphic to $\tilde{X}$ and $G/\tilde{X}$. For any other cycle-separating 4-edge-cut S , one S -factor P contains X as an atom, and a corresponding cut $p_X(S)$ is cycle-separating in $G/\tilde{X}$. The two refinements obtained by using S and the atomic cut in either order give equivalent three-member chains.*

Proof. We follow the cut analysis of [CS10, Proposition 8.2, Lemmas 9.1–9.2, Propositions 9.3–9.4, and Lemma 10.1]. The hypotheses above provide colour-open poles and their completions at every step; each completed graph has girth at least five and cyclic connectivity at least four.

First consider a quasiatomic shore $X \cup \{u, v\}$. Its pole is heterochromatic: an isochromatic completion would place the adjacent vertices u, v beside the two new cap vertices and create a triangle or a quadrilateral. Joining its boundary couples instead completes the pole of X by an adjacent pair. Thus the completed shore is isomorphic to $\tilde{X}$. In the complementary completion, the same isomorphism exchanges the original pair u, v with the added pair and fixes the other vertices. The associated cuts therefore give equivalent factor pairs.

For the remaining cuts, put $T = \partial_G X$. The following two restrictions control the completions:

$$|S \cap T| \neq 3, \quad |S \cap T| = 2 \implies S \cap T \text{ is a couple of neither cut.} \quad (3)$$

For three common edges, the two remaining edges of $S \cup T$ separate a cyclic 2-pole from X , contrary to cyclic 4-edge-connectivity. Here

⁶ Roman Nedela and Martin Škoviera. *Atoms of cyclic connectivity in cubic graphs*. *Mathematica Slovaca* 45(5) (1995), 481–499. DML-CZ:132921.

the 2-pole has vertices, so the cubic degree count forces it to contain a cycle. For two common edges forming an S -couple, complete the S -shore containing X . A heterochromatic completion produces a cycle-separating 2-edge-cut; an isochromatic completion produces an independent 3-edge-cut. Both contradict cyclic 4-edge-connectivity of the completed factor. If the two common edges form a T -couple, restricting colourings of the larger shore pole to the pole of X transfers the corresponding equality or inequality of colours to them. They consequently form an S -couple as well. This proves (3).

The atom theorem puts X inside one S -shore. Its completion P keeps X induced: an added edge within X would join the two common edges as an S -couple, contrary to (3). If $P - X$ is acyclic, the cubic 4-pole degree count leaves two adjacent vertices; reversing the completion identifies S as an atomic or quasiatomic cut. For the cuts under consideration, $P - X$ therefore contains a cycle. Thus X remains a cyclic fragment of P . Its minimality follows because every cyclic fragment properly contained in X in P would also be a cyclic fragment of G . Hence X is an atom of P .

To describe the corresponding cut, replace the pole of X with its colour-equivalent acyclic 4-pole. For a heterochromatic pole, this acyclic pole consists of two adjacent vertices with two semiedges at each; for an isochromatic pole, it consists of two isolated edges. The resulting graph is $G/\tilde{X}$. The map p_X keeps edges outside X and maps the four boundary edges to their replacement edges. In the second case, the two edges of each couple have the same image. By (3), $p_X(S)$ has four distinct edges. The shore of $G - S$ containing X also has vertices outside X , since $S \neq \partial_G X$. After deleting $p_X(S)$, these vertices remain separated from the other shore. Thus $p_X(S)$ is an edge-cut.

The images are also independent. In the adjacent-vertex replacement, an adjacency could arise only from a boundary couple contained in S , which (3) excludes. In the isolated-edge replacement, an inherited edge might become adjacent to a new joined edge. Move their common vertex to the other shore of the projected cut. This gives a 3-edge-cut, which isolates a vertex in the completed snark. The other two projected edges therefore meet there. Their preimages in S are independent, so one is a second boundary edge of X . The resulting configuration consists of X and the adjacent pair of these two outside vertices; its boundary is S . This is precisely the quasiatomic configuration. Thus $p_X(S)$ is independent. Both shores of an independent cut in a cubic graph have minimum degree at least two, and hence contain cycles. This proves the asserted projection property.

Finally, let P, Q be the S -factors, with $X \subseteq P$. Decomposing P at X gives the chain

$$\{\tilde{X}, P/\tilde{X}, Q\}.$$

If the atomic cut is used first, decomposing $G/\tilde{X}$ at $p_X(S)$ leaves the pole defining Q intact. The other pole is obtained from the pole defining P by the colour-equivalent replacement of X . Colour-equivalent substitution preserves its boundary colourings, so its canonical completion is $P/\tilde{X}$. The two chains are equivalent. $\square$

Lemma 4.2. *For $G \in \mathcal{C}$ of cyclic connectivity four and an atom X , every terminal chain of G is equivalent to the union of a terminal chain of $\tilde{X}$ and a terminal chain of $G/\tilde{X}$.*

Proof. Follow X through a terminal chain. By lemma 4.1, each cut outside the associated cuts preserves X as an atom in one factor. Eventually an associated cut is used, since a terminal factor has cyclic connectivity at least five. Replace that step with the atomic cut using the first assertion of lemma 4.1.

Move the atomic step towards the start of the chain. It commutes with a preceding step in a different factor. When the preceding step created the factor containing X , the last assertion of lemma 4.1 interchanges these two steps and preserves the resulting three factors up to isomorphism. Transport the later refinements across these isomorphisms. Each interchange moves the atomic step one position earlier, so finitely many interchanges place it first. The subsequent steps refine its two factors, yielding the two terminal chains in the statement. This is the splitting argument of [CS10, Lemma 10.2]. $\square$

Theorem 4.3. *Let $\mathcal{C}$ be a class of snarks closed under isomorphism. Suppose that the shore poles of every cycle-separating 4-edge-cut in a member of $\mathcal{C}$ are colourable, and that their canonical snark completions belong to $\mathcal{C}$. Then every member of $\mathcal{C}$ has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected factors in $\mathcal{C}$.*

Proof. A canonical decomposition has two factors in $\mathcal{C}$ satisfying (2). Each has at least ten vertices, so both are smaller than the parent. Thus every chain has a terminal refinement, and the terminal members have cyclic connectivity at least five.

For uniqueness, use induction on $|G|$. A cyclically 5-edge-connected member has the single-member terminal chain $\{G\}$. Otherwise choose an atom X and let $\mathcal{F}$ and $\mathcal{F}'$ be two terminal chains. By lemma 4.2,

$$\begin{aligned}\mathcal{F} &\simeq \mathcal{F}_1 \uplus \mathcal{F}_2, \\ \mathcal{F}' &\simeq \mathcal{F}'_1 \uplus \mathcal{F}'_2,\end{aligned}$$

where the first subchains factor $\widetilde{X}$ and the second factor $G/\widetilde{X}$. Both graphs are smaller members of $\mathcal{C}$. The induction hypothesis gives $\mathcal{F}_1 \simeq \mathcal{F}'_1$ and $\mathcal{F}_2 \simeq \mathcal{F}'_2$, and hence $\mathcal{F} \simeq \mathcal{F}'$. $\square$

5 Permutation factors

We verify the two closure hypotheses of theorem 4.3 using a permutation 2-factor. The fixed-cut closure is the result of [MS19, Section 2]; the following proof also describes its colour-open completions.

Lemma 5.1. *Let G be a permutation snark with permutation 2-factor $C_1 \cup C_2$, and let S be a cycle-separating 4-edge-cut. The cut consists of two edges of each C_i . Each shore pole is colourable, and its canonical completion is a permutation snark.*

Proof. Write $M = E(G) \setminus E(C_1 \cup C_2)$ and let the common length of the cycles be m . Each cycle is cut by S . Indeed, if a whole C_i lies in one shore, every vertex in the other shore sends its matching edge across S . That shore then has at most four vertices, and girth at least five makes it acyclic. This contradicts cycle separation. Each cycle contributes a positive even number of cut edges. Thus S has two edges on each cycle, and every edge of M stays within a shore.

Each shore contains one path from each C_i . Its matching edges pair their vertices, so the two paths have the same order r . Colour the matching edges 3 and alternate 1, 2 along each path, including the two boundary semiedges at its ends. The initial colour on each path can be chosen independently. This gives a proper colouring of the shore pole.

The two path ends form the boundary couples. If r is even, the semiedges at the ends of each path receive the same colour. The independent phase choices give types 1111 and 1122, with the two paths defining the pairs. The pole is isochromatic and its completion adds one vertex to close each path, together with a matching edge between the new vertices. If r is odd, the end semiedges receive distinct colours, and the phase choices give types 1212 and 1221. The pole is heterochromatic and its completion joins the ends of each path. In both cases the completed graph has two induced cycles and a perfect matching between them. Its canonical completion has chromatic index four. An uncolourable cycle permutation graph is cyclically 4-edge-connected and has girth at least five [MS19, Section 1]; hence the completed graph is a permutation snark. $\square$

Theorem 1.2. *Every permutation snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected permutation snark factors. Every intermediate factor in a decomposition along cycle-separating 4-edge-cuts is a permutation snark.*

Proof. Apply theorem 4.3 to the class of permutation snarks. By lemma 5.1, every cycle-separating 4-edge-cut has colourable shore poles, and both canonical factors are permutation snarks. The theorem gives existence and uniqueness of the terminal factor multiset. The same closure statement applied at each step keeps all intermediate factors in the permutation class. $\square$

Corollary 1.3. *Every hypohamiltonian permutation snark has a unique multiset, up to isomorphism and with multiplicities, of cyclically 5-edge-connected hypohamiltonian permutation snark factors. Every intermediate factor has both properties.*

Proof. At any cycle-separating 4-edge-cut of a hypohamiltonian permutation snark, lemma 5.1 gives two canonical permutation snark factors. By lemma 3.2, both are hypohamiltonian. This applies at every later step. Existence and uniqueness follow from theorems 1.1 and 1.2. In particular, the hypohamiltonian, permutation and bicritical decompositions of this graph have the same terminal multiset. $\square$

6 Orders of composition factors

The order identity (2) gives a bound on the number of terminal factors and a congruence for their orders.

Corollary 6.1. *Let $H_1, \dots, H_t$ be the terminal factors of a hypohamiltonian or permutation snark G . Then*

$$|G| - 2 = \sum_{i=1}^t (|H_i| - 2), \quad t \leq \frac{|G| - 2}{8}.$$

For a permutation snark, each $|H_i| \equiv 2 \pmod{4}$ and

$$\frac{|G| - 2}{4} \equiv \sum_{i=1}^t \frac{|H_i| - 2}{4} \pmod{2}.$$

Proof. Iterate (2) along the decomposition. Every snark factor has order at least ten, giving the bound on t . For permutation factors, the two induced cycles have equal odd length, so their orders are 2 (mod 4). Dividing the order identity by four gives the congruence. $\square$

Consequently the existence of a permutation snark of order 6 (mod 8) is equivalent to the existence of a cyclically 5-edge-connected permutation snark of that order residue. Indeed, an odd value of $(|G| - 2)/4$ forces an odd contribution from at least one terminal factor.

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